Risk Management as Internal Information Production

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1 Research Question and Motivation

In multidivisional firms, headquarters faces a basic informational problem: divisional performance signals are contaminated by macroeconomic shocks that managers do not control. Exchange rates, commodity prices, freight rates, weather, and other hedgeable risks move budgets, profits, and reported performance in ways that have nothing to do with operating quality. For example, Brown (2001) documents that this contamination is not merely an accounting nuisance. At the multinational firm he studies, hedging is centralized at treasury, internal hedge rates feed directly into divisional budgets and pricing decisions, and divisional managers actively lobby treasury over those rates because they shape both targets and evaluations. Harris and Rajgopal (2022) broaden the empirical picture, showing that internal reporting practices for foreign currency are frequently inconsistent across translation, budgeting, and performance-evaluation systems, and that senior managers and boards review parent-currency numbers that mix operating performance with currency movements in ways that complicate inference about divisional quality.

This proposal develops the hypothesis that derivative markets affect the internal organization of multidivisional firms not primarily through their first-order effect on cash flow or external signaling, but through their second-order effect on the internal information environment in which contracts are written and decision rights are allocated. The central question becomes: when derivative instruments become available for a particular class of operating risk, how do firms reorganize internally? Specifically, how do they adjust contracts, information acquisition, and the allocation of operating authority between headquarters and divisional managers?

The question is significant because the existing literature has examined these margins largely in isolation. Corporate hedging theory analyzes when firms should hedge and what motives drive hedging (Smith and Stulz, 1985; Froot, Scharfstein, and Stein, 1993; DeMarzo and Duffie, 1995), but typically treats the firm as a unitary decision-maker. Contracting theory analyzes how performance signals should be incorporated into compensation under moral hazard (Holmström and Milgrom, 1987; Banker and Datar, 1989; Feltham and Xie, 1994), but rarely connects this to the source of signal noise or to delegation decisions. The literature on delegation and authority analyzes when decision rights should be allocated to subordinates (Aghion and Tirole, 1997; Dessein, 2002), but treats the information environment as a primitive rather than as something firms actively shape.

Recent accounting work has shown that internal information environments matter for organizational design (Ferracuti, 2022; Binz, Ferracuti, and Joos, 2023), but has not identified hedging as a source of internal information production.

The synthesis I propose is that hedging activity, contracts, and decision rights respond *jointly* to a shock in derivative availability, and that the joint response is signed by the correlation between firm exposure and operating productivity. The framework I develop predicts that derivative innovation triggers three distinct forces: contracting raises optimal incentive intensity, information acquisition at divisions responds to higher-powered contracts, and the central treasury's hedging design activity reveals operating structure to headquarters. Whether the net effect on operating authority is centralizing or decentralizing depends on the strength of these forces, which varies across firms in a way that is empirically observable.

I also predict that some firms hedge not only because of the standard motives identified by Smith and Stulz (1985) or Froot, Scharfstein, and Stein (1993), but also because the act of hedging produces internal information useful for managing divisions and stronger incentives for local managers to acquire information. This prediction is not generated by existing hedging theories and can provide new insight into the long standing puzzle that firms often deviate from seemingly optimal hedge ratios.

2 Literature

This proposal sits at the intersection of five literatures: corporate hedging theory, the empirical literature on hedging practices, contracting under noisy performance signals, the theory of delegation and decision rights, and the emerging literature on internal information environments and organizational design. I review each in turn before describing where my proposed model fits.

2.1 Theories of corporate hedging

The modern theory of corporate hedging is built on the observation that, in frictionless markets, hedging by a publicly traded firm is irrelevant: shareholders can replicate any hedge in their personal portfolios. The literature has therefore focused on identifying frictions that make hedging valuable at the firm level. Smith and Stulz (1985) identify three: convex tax schedules (where hedging reduces expected taxes by smoothing pre-tax income), financial distress costs (where hedging reduces the expected costs of approaching distress), and managerial risk aversion (where undiversified managers prefer less volatile compensation). Stulz (1984, 1996) extend the managerial motive analysis and propose that hedging should be understood as managing the lower tail of the cash flow distribution to preserve strategic optionality.

Froot, Scharfstein, and Stein (1993) develop the canonical investment-financing coordination model. When external finance is costly, firms with valuable investment opportunities benefit from stabilizing internal cash flows so that they can fund positive NPV projects without raising external capital at a premium. The optimal hedge ratio depends on the correlation between investment opportunities and hedgeable risks. Gilje and Taillard (2017) provide quasi-experimental evidence consistent with this framework, while Campello, Lin, Ma, and Zou (2011) show that hedging firms face looser capex covenants and invest more, again consistent with the FSS coordination story.

DeMarzo and Duffie (1995) (and their earlier 1991 paper) offer a different rationale: hedging as external information production. In their model, hedging strips noise from the earnings signal that the labor market and shareholders use to infer managerial ability. This generates a signal extraction motive for hedging that depends on the prevailing disclosure regime. Breeden and Viswanathan (1998) extend this to career concerns. The DeMarzo Duffie information channel runs entirely through external markets; the model I propose internalizes this logic, identifying a parallel channel that operates within the firm between headquarters and divisions.

Survey and field study evidence largely supports the standard motives but also documents heterogeneity that is not fully explained by them. Giambona et al. (2018) survey 1100 risk managers and find that 70-80% report hedging to smooth earnings or meet investor expectations, and 90% report hedging to protect cash flow—consistent with FSS and external-signaling motives. However, they also find that firms frequently use operational rather than financial hedging and that contracting choices (OTC vs. exchange-traded) reflect institutional rather than purely economic considerations. Brown (2001) provides a detailed field study at one large multinational, documenting that hedging decisions are deeply embedded in internal control processes: hedge rates are centrally set, monitored, and contested, and they feed directly into divisional budgets, pricing decisions, and performance evaluations. The Brown evidence suggests that hedging functions as an internal organizational tool, not merely a financial risk-management tool, in a way that the canonical theories do not capture.

2.2 Hedging, accounting, and the internal information environment

A second strand of literature examines how hedging interacts with accounting measurement and internal reporting. Harris and Rajgopal (2022) provide a comprehensive review of foreign currency accounting and risk management, documenting that internal practices for translating, budgeting, and evaluating foreign operations are frequently internally inconsistent and that mismatches between translation rates, hedge rates, and operating performance create persistent measurement frictions. They find that translation choices (functional currency designation under ASC 830) are often mechanical and path-dependent, suggesting that the formal accounting structure may not be a margin firms actively optimize.

Bushman, Indjejikian, and Smith (1995) analyze the design of aggregate vs. divisional performance measures in the presence of intrafirm interdependencies, showing that the optimal weighting of measures depends on the structure of the underlying noise. Bushman, Indjejikian, and Smith (1996) extend this to subjective performance evaluation. These models provide the formal apparatus for thinking about how the choice of performance measures responds to changes in the noise environment, but they do not endogenize the noise environment itself.

Ferracuti (2022) shows that public information uncertainty shapes organizational design choices, including hiring and vertical integration. Binz, Ferracuti, and Joos (2023) demonstrate that internal information systems serve as a transmission channel for macroeconomic shocks: firms with better internal systems invest more efficiently in the face of inflation. Ferracuti, Vashishtha, and Wang (2025) examine the closest setting to mine, showing that firms smooth earnings less when noise is hedgeable. Their result is identified on the *external* reporting margin: hedging substitutes for smoothing because both reduce the variance of reported earnings seen by outside investors. My contribution moves to the *internal* margin: hedging changes the noise structure of the divisional signals that headquarters uses to evaluate, compensate, and delegate to managers. The two channels are complementary but operate on different audiences and predict different organizational responses.

Armstrong, Glaeser, and Huang (2022) are the most directly relevant empirical antecedent. They use the introduction of CME exchange-traded weather derivatives as a quasi-experiment and show that, once weather risk becomes contractible (and therefore controllable), executive compensation shifts toward higher-powered incentives for affected firms. This is the Holmström-Milgrom comparative static applied to a derivative-availability shock, and it corresponds to Prediction 1 in my framework. My incremental contribution is to extend the analysis beyond incentive intensity to decision rights and internal information production: the same shock that strengthens incentives can either expand or contract managerial authority, with the cross-sectional direction signed by α.

2.3 Contracting under noisy performance signals

The contracting backbone I borrow is the linear-exponential-normal framework of Holmström and Milgrom (1987). In this framework, a risk-neutral principal contracts with a risk-averse agent under a normally distributed performance signal; the optimal contract is linear in the signal and the optimal incentive intensity is decreasing in signal noise. Holmström (1979) provides the foundation in the informativeness principle: any incrementally informative signal should receive non-zero weight in compensation, regardless of whether the agent controls it.

Holmström and Milgrom (1991) extend the analysis to multitask environments. When agents face multiple unobservable activities and only some are measured, contracts must trade off across tasks; in particular, distorting the relative weight on one task can induce inefficient effort allocation. Banker and Datar (1989) formalize the sensitivity-precision criterion for performance measure design, showing that optimal weights are proportional to the sensitivity-to-precision ratio. Feltham and Xie (1994) introduce the concept of measure congruity in multitask settings. Lambert (2001) provides a comprehensive survey of the contracting framework as applied to accounting research.

In the divisional context, Bushman, Indjejikian, and Smith (1995) analyze how aggregate vs. divisional measures should be combined when divisions are interdependent. Antle and Demski (1988) reinterpret the controllability principle through the informativeness lens: divisional managers should be evaluated on measures that are conditionally informative about their effort, regardless of whether they "control" the underlying outcome. This logic motivates my treatment of hedging as a shock to signal informativeness rather than as a shock to controllability per se.

The relevant comparative static for my purposes is that incentive intensity rises as signal noise falls. Armstrong, Glaeser, and Huang (2022) document this empirically using derivative innovation as the noise shock. I extend their framework by linking the change in incentive intensity to a change in decision rights—hedging affects not just how strongly divisional managers are compensated for performance, but also how much authority they have to act on local information.

2.4 Delegation, decision rights, and the scope of authority

The delegation framework I borrow follows Aghion and Tirole (1997), who distinguish formal authority (the legal right to decide) from real authority (effective control given information). In their model, the principal can acquire information at cost; doing so shifts real authority upward even if formal authority is unchanged. Dessein (2002) adds strategic communication via cheap talk, showing that full delegation can dominate when the preference conflict between principal and agent is small relative to environmental uncertainty. Alonso, Dessein, and Matouschek (2008) extend the analysis to coordination problems in multidivisional firms. Rantakari (2008) examines governance of adaptation. Hart and Holmström (2010) integrate authority with firm scope.

The empirical operationalization of these models has been developed primarily by Bloom, Sadun, and Van Reenen (2012); Bloom, Garicano, Sadun, and Van Reenen (2014) and Acemoglu et al. (2007). Bloom, Sadun, and Van Reenen (2012) use the World Management Survey to construct firm-level decentralization indices and document large cross-country and cross-industry variation in delegation practices. Bloom, Garicano, Sadun, and Van Reenen (2014) sharpen the theory by distinguishing information technology (which empowers subordinates with local data) from communication technology (which enables hard signals to flow upward), and they show that the two types of ICT have opposite effects on decentralization—decentralizing and centralizing, respectively. Acemoglu et al. (2007) use distance-to-frontier and ICT intensity to show that firms in industries where local information is more valuable decentralize more. These empirical templates inform my proposed empirical strategy.

Graham, Harvey, and Puri (2015) survey 1050 CEOs and CFOs and document substantial heterogeneity in delegation practices across decision domains: capital structure decisions are typically centralized, while operational decisions are more often delegated. This heterogeneity is consistent with the cross-sectional pattern my model predicts and provides a natural data source for empirical validation.

Jost and Rohlfing-Bastian (2026) provide a recent theoretical analysis of the scope of delegation in hierarchies. Their setting differs from mine—they consider an agent contracting with a subordinate, which is unusual in corporate hierarchies—but their identification of an "activity-reduction effect," in which an informed party narrows the authorized actions of a subordinate, captures a mechanism that may be relevant in my setting. I borrow the conceptual logic of activity reduction even though I do not adopt their formal structure.

2.5 Hierarchical information acquisition and organizational design

The literature on knowledge hierarchies, beginning with Garicano (2000), provides the conceptual backdrop for thinking about how organizational form responds to changes in the information environment. In Garicano's model, lower layers handle routine problems using local knowledge while higher layers handle exceptions requiring specialized expertise. The equilibrium hierarchy responds to the costs of learning and communicating. Bloom, Garicano, Sadun, and Van Reenen (2014) operationalize this for ICT shocks; the framework would predict that any shock to the cost or feasibility of information acquisition reshapes the hierarchy.

Stein (2002) develops the distinction between hard information (codifiable and transmissible up the hierarchy) and soft information (uncodifiable, requiring local action). Hard information supports hierarchical organization; soft information supports decentralization. The hedge-derived information in my proposed model is paradigmatically hard—it is mark-to-market position data, mandatorily disclosed under FAS 133 / ASC 815, third-party verifiable. This is what enables it to flow up to headquarters and shift the locus of authority. The Stein framework provides the conceptual justification for why F3 (hedge-derived information leakage) actually transmits to headquarters rather than getting trapped at the treasury level.

Diamond (1984) provides the structural template for the bundled information acquisition that drives F3 in my model. In Diamond's model of delegated monitoring, banks acquire information about borrowers as an inseparable byproduct of lending; the information cannot be acquired without the lending activity, and the lending cannot be conducted without acquiring the information. My model adapts this structure: treasury cannot design a hedge without knowing exposure, exposure information is partially informative about operations, and therefore hedging activity produces internal information at headquarters as a byproduct. Rajan (1992) extends the bundling logic to relationship banking, and Petersen and Rajan (1994) provide empirical evidence on the value of relationship-derived information.

Stein (1997, 2002) also analyzes internal capital markets directly, showing that headquarters' ability to engage in winner-picking across divisions depends on the quality of cross-divisional information. Rajan, Servaes, and Zingales (2000) and Scharfstein and Stein (2000) examine the dark side—socialism and rent-seeking in capital allocation—while Ozbas and Scharfstein (2010) provide empirical evidence consistent with these frictions. Wulf (2009) documents influence costs in internal capital markets. The implication for my model is that hedge-derived information at headquarters can affect capital allocation as well as operating authority, though I leave the capital allocation extension for future work.

3 Theoretical Framework

3.1 Intuition: a single shock and three forces

Consider a multinational manufacturer with an Italian subsidiary. The subsidiary's reported EBIT in dollars reflects both operating performance (how well the manager runs the plant, manages suppliers, prices products) and currency movements (the euro-dollar exchange rate over the period). Headquarters in North Carolina uses these reports to evaluate the manager, set bonuses, and decide how much capital to allocate to the subsidiary next year.

Before currency derivatives are available, the reports are noisy. Headquarters cannot easily separate operating performance from currency luck. Three consequences follow. First, contracts tying the manager's pay to reported EBIT must be low-powered since a strongly performance-tied bonus would impose too much FX risk on the risk-averse manager. Second, the manager has limited incentive to invest in local information (about supplier reliability, customer needs, pricing strategy) because that information is not strongly reflected in her compensation. Third, headquarters is reluctant to delegate operating decisions because it cannot tell, from the reports, whether delegated decisions are being made well.

When currency derivatives become available, three things change. The signal becomes cleaner, so contracts can become higher-powered. Higher-powered contracts give the manager stronger incentives to acquire and use local information. Together, these effects increase the value of delegating operating authority to the manager. I refer to these forces as **F1+F2**: enabling incentivized information acquisition and supporting delegation.

But hedging also has a second effect that pulls in the opposite direction. To design a currency hedge, treasury must know the subsidiary's exposure: which currencies appear in revenue, what notional amounts, with what timing. This information is partially informative about the subsidiary's operating structure: a subsidiary that hedges the euro revenue tells treasury something about which markets the subsidiary serves and how it sources inputs. Once treasury has this information, headquarters has a new channel for learning about the subsidiary that bypasses the noisy reports. I call this **F3**: hedge-derived information acquisition by headquarters, supporting retention of authority.

A natural question is why headquarters cannot simply acquire exposure information directly, without going through a hedging activity. The model takes the position that pre-derivative information acquisition by P is prohibitively costly while post-derivative acquisition is approximately free, and that this asymmetry is what gives derivative innovation organizational bite. Three considerations support this asymmetry.

First, similar to the ICT logic, where a technology shock changes both what is possible and what is worth investing in to exploit it (Acemoglu et al., 2007), HQ could in principle acquire information about exposure structure, but the institutional capacity to do so systematically is built around the hedging activity. When hedging becomes available, the treasury function, exposure-mapping systems, and reporting infrastructure get built; without hedging, the infrastructure is not worth building.

Second, direct investigation by HQ is costly both directly (audit fees, executive time) and indirectly (it signals distrust, depresses A's morale, and may erode her effort and information acquisition by signaling that HQ does not value local autonomy). Pre-derivative, the cost-benefit calculus supports limited direct acquisition. Post-derivative, hedging happens for non-information reasons: cash flow smoothing, tax convexity, distress costs, regulatory capital relief, and signaling to creditors. These motives drive the hedging decision. Treasury must elicit exposure information to execute the hedge, but this elicitation is happening anyway, as part of an activity the firm wants to do for other reasons. The marginal cost of the byproduct information is therefore approximately zero.

Third, and most importantly, the asymmetry is reinforced by the agent's incentive to reveal λ truthfully post-derivative but not pre-derivative. Pre-derivative, the agent has no operational reason to disclose her exposure structure: doing so would only reduce her informational rents over autonomy, capital, and compensation. Post-derivative, A herself benefits from accurate hedging because the hedge reduces the variance of her β-linked compensation; if she under-reports λ, treasury hedges less and she bears residual variance that she would prefer to eliminate. Moreover, the realized hedge effectiveness is observable ex post, so misreporting is detectable. Together, these features make truthful disclosure incentive compatible post-derivative in a way that it is not pre-derivative. I formalize this asymmetry in Section 3.3 and Appendix A by treating P as observing λ only when $h = 1$.

The two channels operate on the same shock but pull authority in opposite directions. Whether the net effect is to expand or contract operating delegation depends on which channel is stronger. The strength of F1+F2 depends on how much the contract can be intensified (the size of the variance reduction) and on the productivity of the manager's information effort. The strength of F3 depends on how informative exposure is about operations, which varies systematically across industries. In a multinational manufacturer whose production location determines both currency exposure and unit costs, F3 is strong: treasury learns a lot about operations from exposure. In a firm with translation exposure on remitted earnings unrelated to operating choices, F3 is weak: treasury learns little.

In short, the cross-sectional response of operating authority to derivative innovation depends on the structural correlation between firm exposure and operating productivity. Firms where exposure and operations are tightly linked should respond to derivative innovation by centralizing operating authority; firms where they are weakly linked should respond by decentralizing. The same availability can affect delegation designs, in opposite directions, depending on a measurable industry level primitive.

3.2 Setup

I formalize this intuition in a one-period game between a principal P (headquarters, including the treasury function) and an agent A (division manager). P is risk-neutral and acts as residual claimant. A is risk-averse with CARA preferences and reservation utility normalized to zero.

The relevant state of the world has two components: local productivity θ_L, observed only by A; and a specialized state θ_S, observed only by P. The firm faces an exposure λ to a hedgeable macro shock, where exposure is correlated with local productivity through a structural parameter α that varies across firms. When α is large, operating choices determine exposure and vice versa; when α is small, exposure is largely independent of operations.

A chooses two efforts: operational effort e (running the division) and information effort η (using local knowledge θ_L to make decisions). The operating decision is either delegated to A (with the resulting match value determined by η) or retained by P (with match value determined by P's information set). Throughout, I treat the operating decision as a function of the decision-maker's information set rather than as a separate choice variable; the expected match value m captures the value of an optimal action conditional on that information. P chooses three things: the contract $w = \alpha_w + \beta y$ paid to A, the delegation indicator $\delta \in \{0, 1\}$, and the hedging decision $h \in \{0, 1\}$.

Hedging has two effects. First, it reduces the variance of the noise in the divisional output signal y, from $\bar{\sigma}_h^2$ when $h = 0$ to $\underline{\sigma}_h^2$ when $h = 1$. Second, it generates information leakage to headquarters: when $h = 1$, treasury must observe exposure λ, and observing λ tightens P's posterior over θ_L by a factor $\kappa(\alpha) \in [0, 1)$. The leakage strength κ is increasing in α.

Pre-derivative, $h = 0$ is forced; the firm cannot hedge. Post-derivative, $h \in \{0, 1\}$ is feasible and P chooses optimally.

3.3 Microfoundations

Each component of the setup is borrowed from a specific theoretical tradition, and the borrowed machinery is what allows the model to be solved in closed form.

Preferences and contracts. The CARA-Normal preference structure and the restriction to linear contracts follow Holmström and Milgrom (1987). Under their continuous-time foundations, linear contracts are optimal and the expected utility of a normally distributed wage can be written as a certainty equivalent (mean minus risk-aversion-weighted variance). This is what makes the principal's optimization tractable.

State variables. The two-component state structure follows Bushman, Indjejikian, and Smith (1995), who model divisional information as a combination of local information (the agent's specialty) and aggregate information (the principal's specialty). This is the standard accounting theory formulation of the divisional contracting problem.

Exposure as a correlated side variable. The treatment of exposure λ as a normally distributed signal correlated with the underlying state θ_L follows the multivariate disclosure literature, particularly Goldstein and Yang (2017). When one variable is observed, beliefs about correlated unobserved variables update via Bayes' rule. The strength of the update is the signal-to-noise ratio $\kappa(\alpha) = \alpha^2\sigma_L^2/(\alpha^2\sigma_L^2 + \sigma_\nu^2)$.

Information effort and decision quality. The treatment of information effort η as a costly activity that produces match value follows the multitask-agency literature (Holmström and Milgrom, 1991; Feltham and Xie, 1994). In my model, information effort is productive only when the agent has decision rights, representing a feature that captures the complementarity between contracts and authority. Without delegation, the agent has no margin to apply information, so the contract cannot incentivize information effort.

Hedging as a bundled activity. The structural feature that hedging requires exposure information as an input and therefore generates information acquisition as a byproduct, follows Diamond (1984). In Diamond's model, banks cannot lend without monitoring; the two activities are bundled. In my model, treasury cannot hedge without eliciting exposure; the two are similarly bundled. This is what makes hedging a source of internal information production rather than just an external risk management tool.

Information transmission to headquarters. The assumption that hedge derived information actually reaches headquarters (rather than being trapped at the treasury level) follows Stein (2002). Hedge positions are paradigmatically hard information: mark-to-market, mandatorily disclosed under FAS 133 / ASC 815, third-party verifiable. Hard information transmits up the hierarchy without communication frictions; soft information does not. The institutional structure of derivative markets ensures that hedge-derived information is hard.

Information acquisition asymmetry. The assumption that P observes the exposure signal λ only when $h = 1$, rather than at all times, reflects the institutional asymmetry developed in Section 3.1. Pre-derivative, no operational reason exists for A to disclose λ, and direct investigation by P is costly enough (in audit cost, executive time, and trust-erosion) that the firm does not undertake it; pre-derivative information acquisition cost C_I^{pre} is effectively prohibitive. Post-derivative, treasury must elicit λ to design the hedge regardless of any organizational motive, so the byproduct cost $C_I^{post} \approx 0$. The model captures this asymmetry by indexing the leakage $\kappa(\alpha)\mu_L$ to h.

Incentive-compatible disclosure. Given that elicitation occurs when $h = 1$, the agent's report of λ is truthful in equilibrium for two reinforcing reasons. First, the hedge reduces the variance of A's β-linked compensation, so A strictly prefers a hedge that matches her actual exposure; underreporting λ would leave residual variance that she bears. Second, hedge effectiveness is observable ex post through mandatorily disclosed mark-to-market positions, which disciplines reporting. This is a stronger condition than Crawford and Sobel (1982) cheap-talk truth-telling because reporting is verifiable; the sender's preferences and the institutional disclosure structure jointly make truthful disclosure incentive-compatible.

Derivative availability as exogenous shock. The treatment of derivative availability as an exogenous parameter shock follows the empirical tradition of Hoberg and Moon (2017) and Armstrong, Glaeser, and Huang (2022). Exchanges launch contracts based on aggregate demand uncorrelated with individual firms' organizational structure, providing identification through institutional variation.

3.4 Equilibrium regimes

The discrete delegation choice implies four possible regimes rather than three. The fourth regime is important because, before derivative availability, the firm may still delegate even if it cannot hedge. The feasible regime set is:

- **Regime A:** $h = 0$, $\delta = 0$**.** No hedge and centralize. Headquarters retains authority, and the divisional performance signal remains noisy.
- **Regime D:** $h = 0$, $\delta = 1$**.** No hedge but delegate. The division receives operating authority even though the performance signal remains contaminated by hedgeable noise. This regime captures firms that delegate because local information is sufficiently productive even without a hedging technology.
- **Regime B:** $h = 1$, $\delta = 1$**.** Hedge and delegate. The firm hedges to clean the divisional signal, writes a higher-powered contract, and delegates operating decisions to the manager. The F1+F2 forces dominate.
- **Regime C:** $h = 1$, $\delta = 0$**.** Hedge and centralize. The firm hedges, but headquarters retains operating authority because the F3 leakage channel makes headquarters' own information channel sufficiently strong.

Pre-derivative, hedging is infeasible, so the firm chooses between Regime A and Regime D. Post-derivative, all four regimes are feasible, and the firm selects the regime that maximizes its objective. Derivative availability can therefore generate three types of organizational responses: no organizational change, a move toward delegation, or a move toward centralization. The cross-sectional pattern is governed by the operations-exposure correlation α, the information productivity index $\rho \equiv \mu_L^2/\phi$, the post-derivative signal variance $\underline{\sigma}^2$, and the hedge cost $\bar{C}$.

3.5 Main results

The formal derivations of the four regimes, including closed-form expressions for the optimal contract intensity in each, are in Appendix A. The main results are summarized in three propositions.

Proposition 1 (Incentive intensity)**.** *Delegation and hedging both raise optimal incentive intensity, holding the other margin fixed:*

$$\beta_D^* > \beta_A^*, \qquad \beta_B^* > \beta_C^*, \qquad \beta_C^* > \beta_A^*, \qquad \beta_B^* > \beta_D^*.$$

Thus, hedge-and-delegate (Regime B) has the highest incentive intensity and no-hedge-and-centralize (Regime A) has the lowest. The relative ranking of no-hedge-but-delegate (Regime D) and hedge-and-centralize (Regime C) is ambiguous and depends on whether the delegation information-productivity effect or the hedging precision effect is stronger.

Proposition 2 (Cross-sectional regime selection among hedging firms)**.** *Among firms that hedge post-derivative ($h = 1$ in equilibrium), Regime B is selected over Regime C iff*

$$\Delta_{BC}(\rho, \underline{\sigma}^2) \equiv \frac{1}{2}\left[\frac{(1+\rho)^2}{1+\rho+r\underline{\sigma}^2} - \frac{1}{1+r\underline{\sigma}^2}\right] > \mu_S + \kappa(\alpha)\mu_L.$$

The left-hand side is the incremental surplus from delegating after hedging; it is increasing in information productivity ρ and in hedging effectiveness, equivalently decreasing in the post-hedge variance $\underline{\sigma}^2$. The right-hand side is the F3-augmented value of retention and is increasing in the operations-exposure correlation α. Therefore, the threshold α^* above which firms hedge and centralize is increasing in ρ and in hedging effectiveness.

Proposition 3 (Internal-information motive for hedging)**.** *There exists a parameter region in which the firm selects Regime C: it hedges and centralizes. In this region, the firm would not hedge for variance reduction alone, but hedges once the F3 information benefit is included. A sufficient condition is*

$$\frac{1}{2(1+r\underline{\sigma}^2)} - \frac{1}{2(1+r\bar{\sigma}^2)} < \bar{C}$$

while

$$\kappa(\alpha)\mu_L + \frac{1}{2(1+r\underline{\sigma}^2)} - \frac{1}{2(1+r\bar{\sigma}^2)} > \bar{C},$$

together with $V_C^* > V_B^*$ and $V_C^* > V_D^*$. The share of firms hedging for this internal-information reason is increasing in α.

Proposition 3 generates the most distinctive empirical prediction: among firms in industries where exposure is highly informative about operations, observed hedging activity should exceed what is predicted by standard hedging motives. The residual is attributable to the F3 motive only when it is accompanied by evidence of stronger HQ information production or greater centralization.

3.6 Cross-sectional implications

The model generates four empirical predictions. Prediction 1 corresponds to existing work and serves as a consistency check; Prediction 2 builds on it; Predictions 3 and 4 are the distinctive cross-sectional implications that separate this framework from prior theories of hedging or delegation.

1. **Incentive intensity rises post-derivative** for firms that hedge. This is the standard Holmström and Milgrom (1987) effect, consistent with Armstrong, Glaeser, and Huang (2022). I treat it as a consistency check.

2. **Information acquisition activity rises post-derivative** primarily for firms that select Regime B (hedge and delegate). The most feasible proxies are divisional or segment reporting granularity, segment-level disclosure detail, finance and analytics job postings, and textual evidence of operating-level forecasting and planning systems.

3. **Operating authority responds heterogeneously across firms**, with the cross-sectional pattern signed by α. Low-α firms are more likely to move toward hedge-and-delegate; high-α firms are more likely to move toward hedge-and-centralize. In the discrete model, the sharp prediction is regime switching rather than continuous narrowing of authority.

4. **Hedging activity contains a component not explained by standard motives**, with this residual increasing in α. This residual should be strongest where hedging is accompanied by greater treasury infrastructure, stronger risk-management disclosure, or more centralized capital allocation. This is the most distinctive prediction of the model: it identifies a hedging motive (internal information production) that is not generated by Smith and Stulz (1985), Froot, Scharfstein, and Stein (1993), or DeMarzo and Duffie (1995), and it is jointly testable with Prediction 3 because the same α moderator governs both.

4 Empirical Strategy

4.1 Identification

I propose to identify the model's predictions using staggered launches of derivative contracts as shocks to derivative availability, following the empirical approach of Hoberg and Moon (2017) and Armstrong, Glaeser, and Huang (2022). Contract launches for currency pairs, commodities, weather indices, and freight indices change the feasibility of hedging specific exposure types. The design is best interpreted as an intent-to-treat test: derivative availability shifts the feasible set, while actual hedging adoption remains an endogenous organizational choice and, in some tests, an outcome.

The empirical specification should compare firms with meaningful pre-existing exposure to a newly hedgeable risk against otherwise similar firms without that exposure, before and after the relevant contract launch. The baseline design is:

$$Y_{it} = \beta_1 PostLaunch_{it} + \gamma_i + \tau_t + X_{it}'\Gamma + \varepsilon_{it},$$

where $PostLaunch_{it}$ indicates that a derivative contract relevant to firm i's primary exposure type has become available. Because derivative adoption is endogenous, the primary treatment should be availability rather than realized hedge use.

Because the launches are staggered across time and exposure types, two-way fixed-effects estimation of the equation above is known to be biased under treatment-effect heterogeneity (Borusyak, Jaravel, and Spiess, 2024; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021). I will therefore implement a stacked difference-in-differences estimator in the spirit of Cengiz, Dube, Lindner, and Zipperer (2019), constructing event-specific cohorts of treated and not-yet-treated control firms around each launch and stacking these cohorts before estimation. I will also report Sun-Abraham interaction-weighted estimates as a robustness check. Pre-trends will be assessed using event-study plots with leads and lags relative to each launch, and the parallel-trends assumption is the central identifying assumption I will test.

4.2 Measuring the moderator α

The key moderator is not exposure intensity alone. It is the extent to which exposure is informative about operations. In the model, α is the structural loading of exposure on local productivity: $\lambda = \alpha\theta_L + \nu$. The leakage strength $\kappa(\alpha)$ that drives the F3 channel is a monotone transformation of the population correlation $\mathrm{corr}(\lambda, \theta_L)$.

The empirical challenge is that neither θ_L nor the population correlation is directly observable. I propose to approximate α using pre-period sensitivities of operating outcomes to the relevant exposure shock. At the firm or industry level, estimate

$$OperatingPerformance_{jt} = a_j + b_j \, ExposureShock_t + u_{jt}$$

using only pre-launch observations, where $OperatingPerformance_{jt}$ is segment operating margin or a similar operations-side measure that strips out reported earnings volatility unrelated to operations, and define $\alpha_j = |\hat{b}_j|$ or the analogous pre-period covariance. The intuition is that when exposure is structurally tied to operations (high α in the model), operating-side performance covaries with the exposure shock in the data. For currency exposures, $ExposureShock_t$ can be the relevant exchange-rate movement; for commodity exposures, it can be oil, gas, metals, or agricultural price shocks; for weather exposures, it can be temperature or precipitation deviations; and for freight exposures, it can be changes in freight-rate indices.

Two caveats are important. First, the empirical estimand $|\hat{b}_j|$ is a slope, not a correlation, and conflates the structural α with the variance of the exposure shock. I will therefore report a normalized version using the pre-period covariance scaled by exposure-shock volatility, $\widehat{\alpha}_j = \widehat{\mathrm{cov}}(OperatingPerformance_j, ExposureShock)/\widehat{\sigma}(ExposureShock)$, as the preferred specification. Second, the pre-period sensitivity may itself be shaped by operational hedging choices made before exchange-traded contracts became available—a firm that hedges operationally pre-derivative will display attenuated $\widehat{b}_j$ for reasons unrelated to its underlying structural α. I will address this by (i) restricting attention to pre-derivative windows in which operational hedging is observably limited, (ii) controlling for pre-period operational hedging proxies from Hoberg and Moon (2017), and (iii) reporting an instrumented version of $\widehat{\alpha}_j$ that uses industry-level production geography or input-output linkages as plausibly exogenous determinants of the structural exposure-operations correlation.

Text based measures such as the Hoberg and Moon (2017) offshore activity index, energy intensity, or revenue geography concentration remain useful for identifying exposed firms, but they should not be the main α proxy unless they are transformed into a pre-period exposure performance sensitivity. This distinction is central: the theory predicts heterogeneity based on whether exposure reveals operations, not merely based on whether exposure exists.

4.3 Main outcomes

Because divisional contracts and formal delegation thresholds are difficult to observe at scale, the main tests should focus on observable implications of the regime choices.

Risk-management infrastructure. If hedging creates an internal information channel, derivative availability should increase treasury and risk-management infrastructure. Feasible measures include textual mentions of centralized treasury, risk committees, enterprise risk management, hedge policy detail, exposure mapping, and derivative notional disclosure depth. Job postings or LinkedIn/Revelio-style occupational data can be used to measure growth in treasury, risk, FP&A, and analytics positions.

Operating authority and centralization. The model predicts that low-α firms should move toward delegation, while high-α firms should move toward centralization. Feasible textual proxies include references to capital-expenditure approval, corporate approval, delegation of authority, regional autonomy, division presidents, centralized procurement, centralized treasury, and corporate risk committees. Where available, hand-collected capex authorization thresholds and World Management Survey decentralization scores can serve as validation tests rather than the main sample-wide measure.

Internal capital allocation. If F3 strengthens headquarters' information set, high-α firms should display stronger HQ-directed allocation after derivative availability. One feasible test uses Compustat segment data and estimates whether segment investment becomes more sensitive to segment profitability, sales growth, or investment opportunities after derivative availability, especially for high-α firms.

Hedging residual. To test the internal-information motive, first predict hedging intensity using standard determinants from the FSS/Smith-Stulz tradition, including size, leverage, cash-flow volatility, tax convexity, financial constraints, and investment opportunities. The model predicts that residual hedging is increasing in α, especially when accompanied by evidence of treasury infrastructure or centralization.

4.4 Cross-sectional tests

The central test is a triple-difference design:

$$Y_{it} = \beta_1 PostLaunch_{it} + \beta_2 PostLaunch_{it} \times \alpha_i + \gamma_i + \tau_t + X_{it}'\Gamma + \varepsilon_{it}.$$

For decentralization outcomes, the model predicts $\beta_2 < 0$: high-α firms centralize relative to low-α firms after derivative availability. For HQ information-production outcomes, such as treasury infrastructure or risk disclosure detail, the model predicts $\beta_2 > 0$. For segment capital allocation sensitivity, the model predicts stronger post-launch allocation-performance sensitivity in high-α firms.

References

Abernethy, M. A., J. F. M. G. Bouwens, and L. A. G. M. van Lent (2004). "Determinants of Control System Design in Divisionalized Firms." *The Accounting Review* 79(3), 545-570.

Acemoglu, D., P. Aghion, C. Lelarge, J. Van Reenen, and F. Zilibotti (2007). "Technology, Information, and the Decentralization of the Firm." *Quarterly Journal of Economics* 122(4), 1759-1799.

Aghion, P. and J. Tirole (1997). "Formal and Real Authority in Organizations." *Journal of Political Economy* 105(1), 1-29.

Alonso, R., W. Dessein, and N. Matouschek (2008). "When Does Coordination Require Centralization?" *American Economic Review* 98(1), 145-179.

Antle, R. and J. Demski (1988). "The Controllability Principle in Responsibility Accounting." *The Accounting Review* 63(4), 700-718.

Armstrong, C. S., S. A. Glaeser, and S. Huang (2022). "Contracting with Controllable Risk." *The Accounting Review* 97(4), 27-50.

Banker, R. D. and S. M. Datar (1989). "Sensitivity, Precision, and Linear Aggregation of Signals for Performance Evaluation." *Journal of Accounting Research* 27(1), 21-39.

Binz, O., E. Ferracuti, and P. Joos (2023). "Investment, Inflation, and the Role of Internal Information Systems as a Transmission Channel." *Journal of Accounting and Economics*.

Bloom, N., L. Garicano, R. Sadun, and J. Van Reenen (2014). "The Distinct Effects of Information Technology and Communication Technology on Firm Organization." *Management Science* 60(12), 2859-2885.

Bloom, N., R. Sadun, and J. Van Reenen (2012). "The Organization of Firms across Countries." *Quarterly Journal of Economics* 127(4), 1663-1705.

Borusyak, K., X. Jaravel, and J. Spiess (2024). "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies* 91(6), 3253-3285.

Breeden, D. and S. Viswanathan (1998). "Why Do Firms Hedge? An Asymmetric Information Model." Working Paper, Duke University.

Brown, G. W. (2001). "Managing Foreign Exchange Risk with Derivatives." *Journal of Financial Economics* 60(2-3), 401-448.

Bushman, R. M., R. J. Indjejikian, and A. Smith (1995). "Aggregate Performance Measures in Business Unit Manager Compensation: The Role of Intrafirm Interdependencies." *Journal of Accounting Research* 33, 101-128.

Bushman, R. M., R. J. Indjejikian, and A. Smith (1996). "CEO Compensation: The Role of Individual Performance Evaluation." *Journal of Accounting and Economics* 21, 161-193.

Callaway, B. and P. H. C. Sant'Anna (2021). "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225(2), 200-230.

Campello, M., C. Lin, Y. Ma, and H. Zou (2011). "The Real and Financial Implications of Corporate Hedging." *Journal of Finance* 66(5), 1615-1647.

Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics* 134(3), 1405-1454.

Crawford, V. P. and J. Sobel (1982). "Strategic Information Transmission." *Econometrica* 50(6), 1431-1451.

DeMarzo, P. M. and D. Duffie (1995). "Corporate Incentives for Hedging and Hedge Accounting." *Review of Financial Studies* 8(3), 743-771.

Dessein, W. (2002). "Authority and Communication in Organizations." *Review of Economic Studies* 69(4), 811-838.

Diamond, D. W. (1984). "Financial Intermediation and Delegated Monitoring." *Review of Economic Studies* 51(3), 393-414.

Eswar, S. (2017). "Does Hedging Reduce the Cost of Delegation?" SSRN Electronic Journal.

Feltham, G. A. and J. Xie (1994). "Performance Measure Congruity and Diversity in Multi-Task Principal/Agent Relations." *The Accounting Review* 69(3), 429-453.

Ferracuti, E. (2022). "Information Uncertainty and Organizational Design." *Journal of Accounting and Economics* 74(1), 101493.

Ferracuti, E., R. Vashishtha, and S. Wang (2025). "Do Firms Smooth Earnings Less When They Can Hedge Noise Better?" *The Accounting Review*.

Froot, K. A., D. S. Scharfstein, and J. C. Stein (1993). "Risk Management: Coordinating Corporate Investment and Financing Policies." *Journal of Finance* 48(5), 1629-1658.

Garicano, L. (2000). "Hierarchies and the Organization of Knowledge in Production." *Journal of Political Economy* 108(5), 874-904.

Giambona, E., J. R. Graham, C. R. Harvey, and G. M. Bodnar (2018). "The Theory and Practice of Corporate Risk Management: Evidence from the Field." *Financial Management* 47(4), 783-832.

Gilje, E. P. and J. P. Taillard (2017). "Does Hedging Affect Firm Value? Evidence from a Natural Experiment." *Review of Financial Studies* 30(12), 4083-4132.

Goldstein, I. and L. Yang (2017). "Information Disclosure in Financial Markets." *Annual Review of Financial Economics* 9, 101-125.

Goodman-Bacon, A. (2021). "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2), 254-277.

Graham, J. R., C. R. Harvey, and M. Puri (2015). "Capital Allocation and Delegation of Decision-Making Authority within Firms." *Journal of Financial Economics* 115(3), 449-470.

Harris, T. and S. Rajgopal (2022). "Foreign Currency: Accounting, Communication and Management of Risks." *Foundations and Trends in Accounting* 16(3), 184-307.

Hart, O. and B. Holmström (2010). "A Theory of Firm Scope." *Quarterly Journal of Economics* 125(2), 483-513.

Hoberg, G. and S. K. Moon (2017). "Offshore Activities and Financial vs. Operational Hedging." *Journal of Financial Economics* 125(2), 217-244.

Holmström, B. (1979). "Moral Hazard and Observability." *Bell Journal of Economics* 10(1), 74-91.

Holmström, B. and P. Milgrom (1987). "Aggregation and Linearity in the Provision of Intertemporal Incentives." *Econometrica* 55(2), 303-328.

Holmström, B. and P. Milgrom (1991). "Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design." *Journal of Law, Economics, and Organization* 7, 24-52.

Jost, P.-J. and A. Rohlfing-Bastian (2026). "Incentives and the Scope of Delegation in Hierarchies." TRR 266 Working Paper No. 232.

Lambert, R. A. (2001). "Contracting Theory and Accounting." *Journal of Accounting and Economics* 32, 3-87.

Ozbas, O. and D. S. Scharfstein (2010). "Evidence on the Dark Side of Internal Capital Markets." *Review of Financial Studies* 23(2), 581-599.

Petersen, M. A. and R. G. Rajan (1994). "The Benefits of Lending Relationships: Evidence from Small Business Data." *Journal of Finance* 49(1), 3-37.

Rajan, R. G. (1992). "Insiders and Outsiders: The Choice between Informed and Arm's-Length Debt." *Journal of Finance* 47(4), 1367-1400.

Rajan, R., H. Servaes, and L. Zingales (2000). "The Cost of Diversity: The Diversification Discount and Inefficient Investment." *Journal of Finance* 55(1), 35-80.

Rantakari, H. (2008). "Governing Adaptation." *Review of Economic Studies* 75(4), 1257-1285.

Scharfstein, D. S. and J. C. Stein (2000). “The Dark Side of Internal Capital Markets: Divisional Rent-Seeking and Inefficient Investment.” *Journal of Finance* 55(6), 2537-2564.

Smith, C. W. and R. M. Stulz (1985). “The Determinants of Firms’ Hedging Policies.” *Journal of Financial and Quantitative Analysis* 20(4), 391-405.

Stein, J. C. (1997). “Internal Capital Markets and the Competition for Corporate Resources.” *Journal of Finance* 52(1), 111-133.

Stein, J. C. (2002). “Information Production and Capital Allocation: Decentralized versus Hierarchical Firms.” *Journal of Finance* 57(5), 1891-1921.

Stulz, R. M. (1984). “Optimal Hedging Policies.” *Journal of Financial and Quantitative Analysis* 19(2), 127-140.

Stulz, R. M. (1996). “Rethinking Risk Management.” *Journal of Applied Corporate Finance* 9(3), 8-25.

Sun, L. and S. Abraham (2021). “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects.” *Journal of Econometrics* 225(2), 175-199.

Wulf, J. (2009). “Influence and Inefficiency in the Internal Capital Market.” *Journal of Economic Behavior and Organization* 72(1), 305-321.

A Formal Model

This appendix provides the formal derivation of the model summarized in Section 3. I first lay out the precise mathematical setup, then solve the agent's problem, then solve the principal's problem under each regime, and finally compare regimes to derive the propositions.

A.1 Mathematical Setup

Players. Risk-neutral principal P; risk-averse agent A with CARA utility $U(w) = -\exp(-rw)$ and risk-aversion parameter $r > 0$.

State variables. $\theta_L \sim \mathcal{N}(0, \sigma_L^2)$ observed by A; $\theta_S \sim \mathcal{N}(0, \sigma_S^2)$ observed by P; $\theta_L \perp \theta_S$.

Exposure. $\lambda = \alpha\theta_L + \nu$, $\nu \sim \mathcal{N}(0, \sigma_\nu^2)$, $\nu \perp (\theta_L, \theta_S)$.

Decisions. Delegation $\delta \in \{0, 1\}$; hedge $h \in \{0, 1\}$. The operating decision is taken by whoever holds authority (A under $\delta = 1$, P under $\delta = 0$) and is optimal conditional on that party's information set; its expected value is captured by the match value m below.

Efforts. Operational effort $e \geq 0$ at cost $e^2/2$; information effort $\eta \geq 0$ at cost $\phi\eta^2/2$.

Output. $y = m + e + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2(h))$ with

$$\sigma^2(h) = \begin{cases} \bar{\sigma}^2 = \bar{\sigma}_h^2 + \sigma_o^2 & \text{if } h = 0 \\ \underline{\sigma}^2 = \underline{\sigma}_h^2 + \sigma_o^2 & \text{if } h = 1 \end{cases}$$

Match value.

$$m = \begin{cases} \eta\mu_L & \text{if } \delta = 1 \text{ (A handles)} \\ \mu_S + \kappa(\alpha)\mu_L \cdot h & \text{if } \delta = 0 \text{ (P handles)} \end{cases}$$

where $\mu_L \equiv \sigma_L^2$, $\mu_S \equiv \sigma_S^2$, and the leakage strength is

$$\kappa(\alpha) = \frac{\alpha^2\sigma_L^2}{\alpha^2\sigma_L^2 + \sigma_\nu^2}$$

Contract. $w = \alpha_w + \beta y$.

Hedge cost. $\bar{C} \cdot h$.

A.2 Agent's Problem

Under CARA preferences and normally distributed wage, the agent's certainty equivalent is

$$CE_A = E[w] - \frac{r}{2}\text{Var}(w) - \text{cost}_A$$

Given $E[y] = m + e$ and $\text{Var}(y) = \sigma^2(h)$:

$$CE_A = \alpha_w + \beta(m+e) - \frac{e^2}{2} - \nVdash[\delta = 1] \cdot \frac{\phi\eta^2}{2} - \frac{r}{2}\beta^2\sigma^2(h)$$

Effort FOC. For all δ:

$$\frac{\partial CE_A}{\partial e} = \beta - e = 0 \implies e^*(\beta) = \beta$$

Information effort FOC. Under $\delta = 1$:

$$\frac{\partial CE_A}{\partial \eta} = \beta\mu_L - \phi\eta = 0 \implies \eta^*(\beta) = \frac{\beta\mu_L}{\phi}$$

Under $\delta = 0$, the agent has no decision authority, so information effort is wasted: $\eta^*(0) = 0$.

Implied match value at the agent's optimum.

$$m^*(\delta, \beta, h) = \begin{cases} \frac{\beta\mu_L^2}{\phi} & \text{if } \delta = 1 \\ \mu_S + \kappa(\alpha)\mu_L \cdot h & \text{if } \delta = 0 \end{cases}$$

A.3 Principal's Problem

Under CARA-Normal aggregation, the principal's expected payoff after the participation constraint binds is total surplus minus the risk premium:

$$V(\beta, \delta, h) = m^*(\delta, \beta, h) + e^*(\beta) - \frac{e^*(\beta)^2}{2} - \nVdash[\delta = 1]\frac{\phi(\eta^*)^2}{2} - \frac{r}{2}\beta^2\sigma^2(h) - \bar{C}h.$$

Because the discrete model has $h \in \{0, 1\}$ and $\delta \in \{0, 1\}$, there are four regimes.

Regime A ($h = 0, \delta = 0$): no hedge and centralize.

$$V_A(\beta) = \mu_S + \beta - \frac{\beta^2}{2}(1 + r\bar{\sigma}^2).$$

The first-order condition is $1 - \beta(1 + r\bar{\sigma}^2) = 0$, yielding

$$\beta_A^* = \frac{1}{1 + r\bar{\sigma}^2}, \qquad V_A^* = \mu_S + \frac{1}{2(1 + r\bar{\sigma}^2)}.$$

Regime D ($h = 0, \delta = 1$): no hedge but delegate. Under delegation, the agent's information effort creates match value and must be incentivized. Let $\rho \equiv \mu_L^2/\phi$ and $A_L \equiv 1 + \rho$. Then

$$V_D(\beta) = A_L\beta - \frac{\beta^2}{2}(A_L + r\bar{\sigma}^2).$$

The first-order condition is $A_L - \beta(A_L + r\bar{\sigma}^2) = 0$, yielding

$$\beta_D^* = \frac{A_L}{A_L + r\bar{\sigma}^2}, \qquad V_D^* = \frac{A_L^2}{2(A_L + r\bar{\sigma}^2)}.$$

Regime B ($h = 1, \delta = 1$): hedge and delegate.

$$V_B(\beta) = A_L\beta - \frac{\beta^2}{2}(A_L + r\underline{\sigma}^2) - \bar{C}.$$

The first-order condition is $A_L - \beta(A_L + r\underline{\sigma}^2) = 0$, yielding

$$\beta_B^* = \frac{A_L}{A_L + r\underline{\sigma}^2}, \qquad V_B^* = \frac{A_L^2}{2(A_L + r\underline{\sigma}^2)} - \bar{C}.$$

Regime C ($h = 1, \delta = 0$): hedge and centralize.

$$V_C(\beta) = \mu_S + \kappa(\alpha)\mu_L + \beta - \frac{\beta^2}{2}(1 + r\underline{\sigma}^2) - \bar{C}.$$

The first-order condition is $1 - \beta(1 + r\underline{\sigma}^2) = 0$, yielding

$$\beta_C^* = \frac{1}{1 + r\underline{\sigma}^2}, \qquad V_C^* = \mu_S + \kappa(\alpha)\mu_L + \frac{1}{2(1 + r\underline{\sigma}^2)} - \bar{C}.$$

A.4 Regime Comparisons

Proof of Proposition 1. First, holding hedging fixed at $h = 0$, delegation raises incentive intensity:

$$\beta_D^* > \beta_A^* \iff \frac{A_L}{A_L + r\bar{\sigma}^2} > \frac{1}{1 + r\bar{\sigma}^2},$$

which holds because $A_L = 1 + \rho > 1$. Holding hedging fixed at $h = 1$, delegation also raises incentive intensity:

$$\beta_B^* > \beta_C^* \iff \frac{A_L}{A_L + r\underline{\sigma}^2} > \frac{1}{1 + r\underline{\sigma}^2},$$

which again holds because $A_L > 1$.

Next, holding delegation fixed at $\delta = 0$, hedging raises incentive intensity because $\underline{\sigma}^2 < \bar{\sigma}^2$:

$$\beta_C^* = \frac{1}{1 + r\underline{\sigma}^2} > \frac{1}{1 + r\bar{\sigma}^2} = \beta_A^*.$$

Holding delegation fixed at $\delta = 1$, hedging also raises incentive intensity:

$$\beta_B^* = \frac{A_L}{A_L + r\underline{\sigma}^2} > \frac{A_L}{A_L + r\bar{\sigma}^2} = \beta_D^*.$$

Therefore Regime B has the highest incentive intensity and Regime A has the lowest. The ranking of β_C^* and β_D^* is ambiguous. ■

Proof of Proposition 2. Among hedging firms, compare V_B^* and V_C^*:

$$V_B^* - V_C^* = \frac{A_L^2}{2(A_L + r\underline{\sigma}^2)} - \frac{1}{2(1 + r\underline{\sigma}^2)} - \mu_S - \kappa(\alpha)\mu_L.$$

Using $A_L = 1 + \rho$, define

$$\Delta_{BC}(\rho, \underline{\sigma}^2) \equiv \frac{1}{2}\left[\frac{(1+\rho)^2}{1 + \rho + r\underline{\sigma}^2} - \frac{1}{1 + r\underline{\sigma}^2}\right].$$

Equivalently,

$$\Delta_{BC}(\rho, \underline{\sigma}^2) = \frac{\rho\left[1 + \rho + r\underline{\sigma}^2(2+\rho)\right]}{2(1 + \rho + r\underline{\sigma}^2)(1 + r\underline{\sigma}^2)}.$$

Therefore Regime B is selected over Regime C iff

$$\Delta_{BC}(\rho, \underline{\sigma}^2) > \mu_S + \kappa(\alpha)\mu_L.$$

The right-hand side is increasing in α because $\kappa'(\alpha) > 0$. The left-hand side is increasing in ρ and decreasing in $\underline{\sigma}^2$. Thus the threshold α^* above which Regime C dominates is increasing in ρ and in hedging effectiveness, where stronger hedging effectiveness corresponds to a lower post-hedge variance $\underline{\sigma}^2$. ■

Proof of Proposition 3. Variance reduction alone would justify hedge-and-centralize over no-hedge-and-centralize only if

$$\frac{1}{2(1 + r\underline{\sigma}^2)} - \frac{1}{2(1 + r\bar{\sigma}^2)} > \bar{C}.$$

If instead

$$\frac{1}{2(1 + r\underline{\sigma}^2)} - \frac{1}{2(1 + r\bar{\sigma}^2)} < \bar{C},$$

then the standard precision benefit is insufficient by itself. With F3, Regime C dominates Regime A when

$$\kappa(\alpha)\mu_L + \frac{1}{2(1 + r\underline{\sigma}^2)} - \frac{1}{2(1 + r\bar{\sigma}^2)} > \bar{C}.$$

For sufficiently high α, $\kappa(\alpha)$ is large, so this condition can hold even when the risk-reduction term alone is too small.

To ensure that Regime C is the selected post-derivative regime, it must also dominate the delegation regimes. From Proposition 2, Regime C dominates Regime B when

$$\kappa(\alpha)\mu_L > \Delta_{BC}(\rho, \underline{\sigma}^2) - \mu_S.$$

It dominates Regime D when

$$\mu_S + \kappa(\alpha)\mu_L + \frac{1}{2(1 + r\underline{\sigma}^2)} - \bar{C} > \frac{(1+\rho)^2}{2(1 + \rho + r\bar{\sigma}^2)}.$$

Because V_C^* is increasing in $\kappa(\alpha)$ while V_A^*, V_B^*, and V_D^* do not depend on α, there is a nonempty parameter region in which high-α firms hedge and centralize. In that region, hedging is chosen not because variance reduction alone pays for the hedge, but because hedge design also produces hard internal information for headquarters. ■